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Lab1: Intro to Matlab & Numerical Methods for solving Ordinary Differential Equation (ODE)

This tutorial is meant to be interactive... that is *you* should be reading, typing, thinking, and asking questions. There are no source files to share (except for section 4.1 and 4.2), the point is for you to think about and enter the code and see the result. And then, to vary the code and see what changes. This is a more effective strategy than just *reading* in code others have written and running it. A few notes:

- If you are experienced in MATLAB, you can start working on the problems (no need to sit through the Matlab tutorial).
- Please do talk to a neighbor if you're stuck, but write your own code!
- You can do it!

1 Getting Started

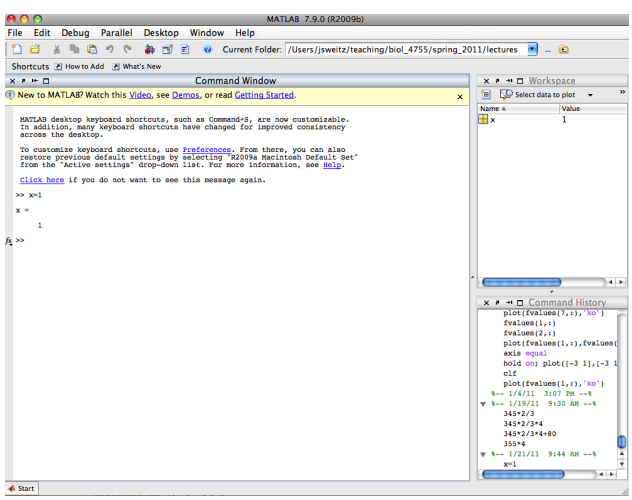
MATLAB is a numerical computing environment and is also a programming language. We will be using it in this class to simulate and model the dynamics of living systems from scales of molecules to ecosystems. MATLAB is particularly good at:

- Matrix manipulation
- Numerical methods
- Algorithms
- Graphical output

and is okay at:

- Symbolic computation
- Interfacing with other languages
- Speed of for loops (as compared to C)

It can also be used for interactive work using the MATLAB Live Editor (similar to the iPython notebook). So, let's get started! In today's class, you will gain practical experiencing using Matlab. When you open Matlab, you should see a set of windows that looks like this:



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The key elements are the Command Window, Workspace and Command History. The command window is where you enter commands, and you should see a prompt that looks like this:

```
>>
```

You can do basic math at this prompt, for example

```
>> 3+4
ans =
     7
```

and

```
>> exp(1)
ans =
    2.7183
```

Matlab has **many** functions that are built in (and you can use it like a calculator), for example, to learn more about the exponential function:

```
>> help exp
 EXP    Exponential.
    EXP(X) is the exponential of the elements of X, e to the X.
    For complex Z=X+i*Y, EXP(Z) = EXP(X)*(COS(Y)+i*SIN(Y)).

    See also expm1, log, log10, expm, expint.

    Overloaded methods:
       lti/exp
       codistributed/exp
       fints/exp

    Reference page in Help browser
       doc exp
```

Many functions have names that you expect (how do you think you should calculate cosine or sine of a value, for example)? Try it out! If you don't know the name of the function you can use the command lookfor or help.

Matlab is not just a calculator. It is also a programming language that can store values in memory. For example, the command

```
>> x=3
x =
     3
```

tells Matlab that the variable x has the value 3 and now every time you use "x", Matlab will substitute the value 3, for example:

```
>> y=x+1
y =
     4
```

It is very important to realize the "=" sign does not mean that Matlab checks to see if the two sides are equal to each other, but rather states that whatever is on the left should be assigned the value of that on the right. If you want to check the truth of a particular statement, for example is x equal to 3, or alternatively, is y equal to 3 then you would type:

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```
>> x==3
ans =
     1
>> y==3
ans =
     0
```

The double "==" sign tells Matlab to logically compare what is on the left with that on the right and return 1 if true and 0 if false. Note that if you don't want Matlab to report back the answer every time, you can use the semicolon

```
>> y=x+1;
```

Matlab can also handle arrays of values (for example a vector or a matrix). The simplest way is to use the colon, which defines a sequential vector, for example

```
>> v=1:5
v =
     1     2     3     4     5
```

you can also modify the increments by adding a step in the middle

```
>> w=1:2:9
w =
     1     3     5     7     9
```

Any entry can be examined using the parentheses

```
>> w(3)
ans =
     5
```

and basic math can be performed automatically on vectors (and matrices), for example

```
>> log(2*w)
ans =
    0.6931    1.7918    2.3026    2.6391    2.8904
```

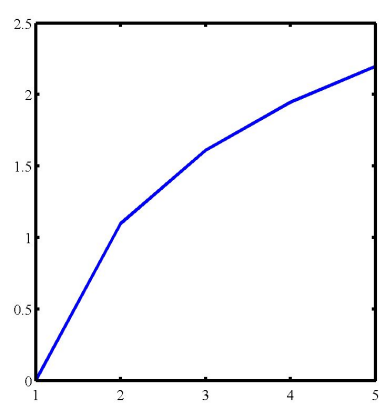
and elementwise operation (denoted by (.) in Matlab) on vectors as well

```
>> w.^2
ans =
     1     9    25    49    81
```

Matlab can also plot graphs and surfaces and all sorts of things. To create a simple plot, use the plot command

```
>> plot(v,log(w))
```

which leads to:



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2 Building "Programs" from "Scripts" and "Functions"

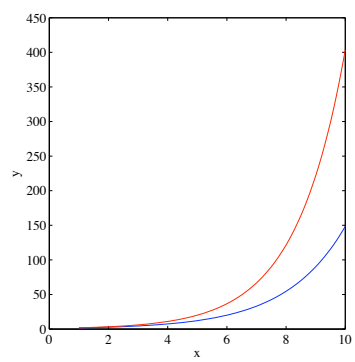
However, once you have a lot of commands, it will get exhausting typing them (especially when you make mistakes). So, Matlab has a concept of a script, which is a list of commands in a file that you can execute directly from the command window. To create a script go to the File menu and select New > M-file. Now wrote a few commands, such as:

```
% my_first_file.m
% Create some vectors
x =1:0.1:10;
y1 =exp(0.5*x);
y2 =exp(0.6*x);
% Plot the vectors
figure  %Pullupanewfigure
hold on %Allow multiple plots on the same figure
plot(x,y1)
plot(x,y2,'r') % Use a red line
xlabel('x'), ylabel('y')  %Label the axes
print -dpdf my_first.pdf % Save the image to a file
```

Save this file as my_first_file.m in the same folder you're currently working in. Now go to the Matlab prompt and type

```
>> my_first_file
```

and it should do all the commands in your script give you your figure.



The problem with this script is that if you wanted to change the exponents in these files, you would have to edit the script and then re-run it, instead of designating a variable change from the command window. Moreover, we cannot have a script return a variable or accept a variable as input. To do, we need to use a concept called a "function". Functions are program files that can be called from the Command window - open a new M-file and type:

```
function dNdt=logGrowth(t,N)
% function dNdt=logGrowth(t,N)
%
% logGrowth gives the growth rate of a population of size N at time t
% Usage:dNdt=logGrowth(t,N)
r =0.5;
K =100;
dNdt =r*N.*(1-N/K);
```

Now you can use your function like one of Matlab's, for example:

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```
>> vN =0:110;
>> figure
>> plot(vN, logGrowth(0,vN))
>> xlabel('N'),ylabel('dN/dt')
```

gives an upside down parabola, denoting that growth rate is positive between 0 and 100 and negative when N is greater than 100.

3 Getting Started with Core Techniques

3.1 Create loops

The two types of loops we will discuss are "for" and "while" loops. Both loop start with a keyword such as for or while and they end with the word end. The "for" loop allows us to repeat certain commands many times with a "counter" variable. Here is one example:

```
>> for j=1:4 % counting
     j+2 % the statement you want to repeat
   end
```

You can also increment your counter variable by any real number like

```
>> for j=1:-0.1:0 % counting
     1-j % the statement you want to repeat
   end
```

Or you can even use a set of arbitrary values:

```
>> for j=[1 3 5 7]
     j-1 % the statement you want to repeat
   end
```

1. Exercise on matrices (10pt). Define a random matrix A of size 3-by-3. Use a double "for loop" to calculate the square of the entries in A and store the values in another matrix B (Hint: type "help rand" in Matlab if you don't know how to define a random matrix). Your solution should show both A and B.

The second type of loop is the "while" loop. The "while" loop repeats a sequence of commands as long as some condition is met. For example, given a number n, the following code will return the smallest non-negative integer a such that $2^a \geq n$.

```
function a = smallexp(n)

a = 0;
while 2^a < n
  a = a + 1; % statement to execute if the condition is met
end
  a = smallexp(4)

a =
     2
```

Note that in the above example we used the conditional statement, $2^a < n$ to decide whether the statement within the while loop should be proceeded. Such conditional statements are also used in "if" statements that are discussed in the next section. The relational operator used here is "<", which means "less than." Other relational operators that are available in Matlab include:

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```
>  greater than
<= less than or equal
>= greater than or equal
== equal
~= not equal
```

Simple conditional statements can be combined by logical operators

```
(&&, ||, ~)
```

into compound expressing such as the following:

```
(5 > 1) && (5 == 6)
```

3.2 Make decisions

Now, let's suppose you want your code to make a decision, and the "if" statement is what you need. The general form in Matlab is as follows:

```
if expression1
  statements1
elseif expression2
  statements2
else
  statements3
end
```

2. Exercise (15pt): Finish the following recursive code (fill in the # part) that gives the factorial of a positive number n, using the recursive formula $n! = n(n-1)...1$. Then test 2 input values of your choice using the code.

```
function N = factorial_recur(n)

if #
    N = 1;
else
    N = #;
end
```

3.3 Go fast, i.e., "vectorize"

Remember: for loops are SLOW in Matlab. One way to make your Matlab run faster is to 'vectorize' the algorithm you use in the code. Vectorization can be done by converting for and while loops to equivalent vector or matrix operations. A simple example would be the following (tic and toc are used as a stop watch):

```
x = 1;
tic
for k = 1:1001
   y(k) = log10(x);
   x = x + .01;
end
toc

tic
x = .01:.01:10;
y = log10(x);
toc
```

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3. Exercise (15pt): Re-do the for loop exercise (Ex.1) using element-wise operator (.$\hat{}$) and see how much time you've saved (hint: use tic, toc function above to measure time).

However, for more complicated code, vectorization is not always so obvious. Some of the most commonly used functions for vectorizing can be found in the Help browser of Matlab.

3.4 Find values you want to know about

Given an array X, the command

```
>> find(X)
```

returns the indices of all nonzero elements of X. You can also use a logical expression to define X. For example,

```
>> find(X>2)
```

returns the indices corresponding to the entries of X that are greater than 2. Notice that X could also be a matrix. In this case, the function returns the linear indices as if the matrix was stored as a single column of elements (Try it!).

However, when you want to locate the entries that satisfy more than one logical expressions, the element-wise logical operators "and" and "or" (denoted by & and | in Matlab, respectively) are used (Can also check out short-circuit logical operators (&& and ||) if you're interested).

4. Exercise(15pt): Build a 5-by-5 random matrix A using the command

```
>> A = rand(5)
```

Find the row and column indices of entries whose value is smaller than 1/4 and bigger than 1/6 (hint: type "help find" to learn more about find function if necessary. Can also use `ind2sub` function to convert linear index to matrix index).

3.5 Save and load data

The keywords in Matlab are straightforward: `save` and `load`. Type

```
>> save filename
```

and Matlab will save all variables in the current workspace in the file filename. If you do not specify an extension to the filename, Matlab uses .mat. When you want to load all the variables from the file specified by filename, just type

```
>> load filename
```

Please keep in mind that these file formats are binary, i.e., not human-readable. If you want to save a particular file format use the help to save variables in standard, comma-separated value file format. There are also alternative ways to load and print data.

4 Numerically Integrating Differential Equations

In this class we can use Matlab to numerically solve differential equations, like the logistic growth equation, even when such solutions are not available analytically. The most-used Matlab program which does the integration is called ode45. Here is a script that integrates the logGrowth function. We will return to this multiple times in the course.

```
% Numerical solution of the logistic equation
t0 =0; % Initial time
tf =50; %Final time
N0 =1; %Initial population size
```

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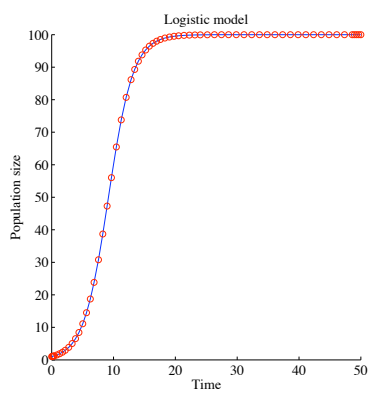
```
[T, vNint] = ode45(@logGrowth, [t0 tf], N0); %Numerically integrate
% Actual solution
r =0.5;
K =100;
vNact =(N0*exp(r*T))./(1+N0*(exp(r*T)-1)/K); %Actual solution
% Plot results
figure;
hold on
plot(T,vNint) % Plot numerically intregrated solution
plot(T,vNact,'ro') %Plot actual solution
xlabel('Time'), ylabel('Population size'),title('Logistic model')
print -dpdf logPlot.pdf
```

Save this script as intLog.m and run it

```
>> intLog
```

and then get the plot



Here are some important points to keep in mind:

- Matlab solves ordinary differential equations of the form

$$\frac{d\vec{y}}{dt} = \vec{f}(\vec{y})$$

  where $\vec{f}(\vec{y})$ is a vector of functions.

- Main solver you should use is: **ode45**.

$$[\text{TOUT,YOUT}] = \text{ODE45(ODEFUN,TSPAN,Y0)}$$

  1. TOUT $\rightarrow$ time of output
  2. YOUT $\rightarrow$ value of variable
  3. ODEFUN $\rightarrow$ name of function containing the dynamics
  4. TSPAN $\rightarrow$ time limits for integration
  5. Y0 $\rightarrow$ initial conditions

- help ode45 for more information

5. Exercise (15pt): read the ode45 documentation and modify your logistic growth model to accept r and K as parameters. Vary the value of K over 1 order of magnitude and compare the results. You should generate 2 plots with different K values and explain your findings.

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4.1 Euler's method

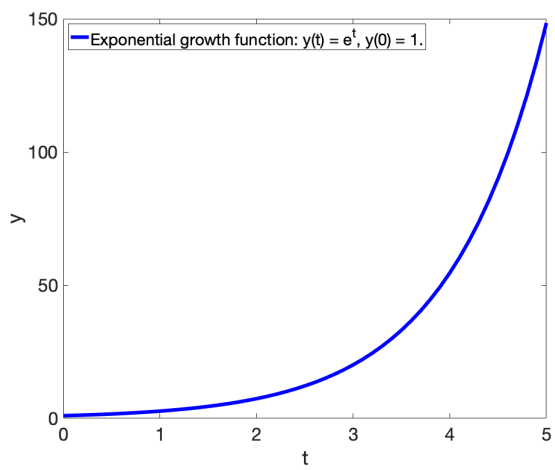
Euler's Method is one of the most well known method for numerically solving first order first degree differential equation that has a given initial condition. Let's understand the method better through an example:

First consider,

$$\frac{dy}{dt} = y \tag{1}$$

$$y(0) = 1 \tag{2}$$

From lecture we know that this is the ode for exponential growth function, and the solution is $y(t) = e^t$. We can plot it for $t \in [0, 5]$ in Matlab, which outputs:



It's great that we can solve the ode analytically, however, it is almost impossible to solve ode systems analytically in real world situation. So do we give up? NO! We can still solve them numerically. In this case, let's use Euler's method to solve this ode numerically. First let's define the function:

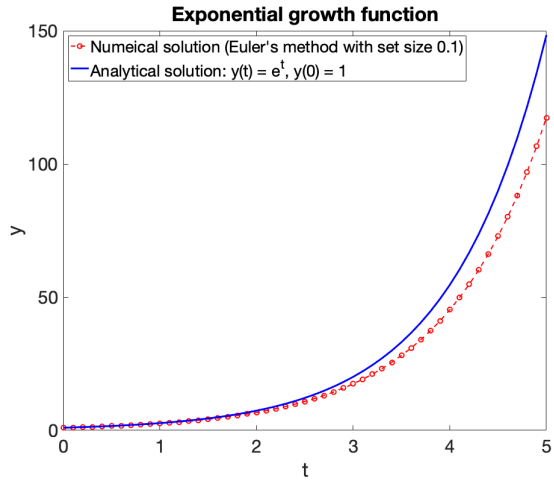
```
function dydt = exponential_growth(t,y)
dydt = y;
```

Then let's solve it using `eul.m` (Euler's method implementation in Matlab) with step size 0.1 and compare it with the analytical solution:

```
t0 = 0;          %Set initial time
tend = 5;          %Set end time
h = 0.1;          %Set time step to 0.1
y0 = 1;          %Set initial condition to y(0) = 1
out = eul(@exponential_growth,t0,h,tend,y0);          % Solve numerically here.
```

Now we can plot and compare:

```
t = 0:0.1:5;                      %define another time here for plotting.
figure
plot(t,out,'r--o',t,exp(t),'b-')
```



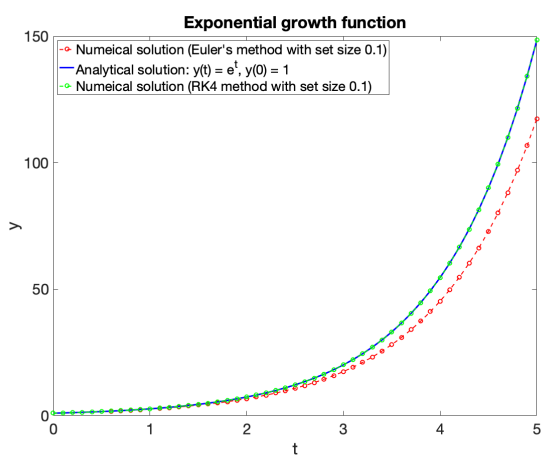
We can see from the plot that Euler's method is using the initial condition and the slope to estimate the solution. It first starts with the initial condition, then it looks at the next time step, which is 0.1 based on our selection of time step. We know that the slope at t = 0 is 1 (based on our initial condition). Then we linearly approximate the y value for next time step 0.1 using the slope.

4.2 Runge-Kutta 4th order method (RK4)

The RK4 is similar to Euler's method, it also uses the initial condition and the slope to estimate the solution, but it uses the **average** slope from one time step to the next to get a better estimate of the solution. The code for solving ode using RK4 is similar to Euler's method. We now use `RK4.m` instead of `eul.m`:

```
t0 = 0;          %Set initial time
tend = 5;          %Set end time
h = 0.1;          %Set time step to 0.1
y0 = 1;          %Set initial condition to y(0) = 1
out = RK4(@exponential_growth,t0,h,tend,y0);          % Solve numerically here.
```

We can plot and compare all three solutions:



We can see that RK4 performs a lot better compare to Euler's method.

6. Exercise (30pt): Given the ode and initial condition:

$$\frac{dy}{dt} = cos(20t) \tag{3}$$

$$y(0) = 1 \tag{4}$$

- Solve for y(t) analytically using `dsolve()` in Matlab, and plot the solution over (0,10) with step size $\triangle$t = 0.01.
- Program the ODE in Matlab and solve with step size $\triangle$t = 0.01, 0.1, and 1 (where needed) over (0,10) using (**3 plots EACH for Euler's method and Runge-Kutta 4th order. One for each step size.**):
  - Euler's method
  - Runge-Kutta 4th order
- For each method and step size, plot the error between analytical and numerical solutions (error means the analytical solution output minus numerical solution output (or vise versa)). Calculate max error over (0,10). Explain the relationship between step size and error for each method.

5 Advanced - Individual-Based Simulations

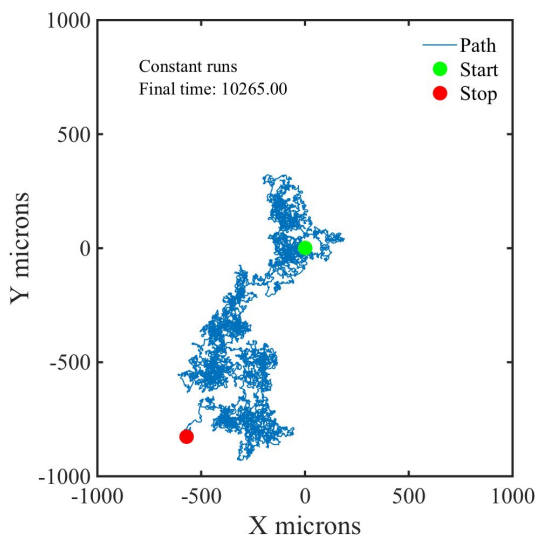
The questions in this section are optional, I'll give out 15 extra points if you can answer the questions in this section correctly

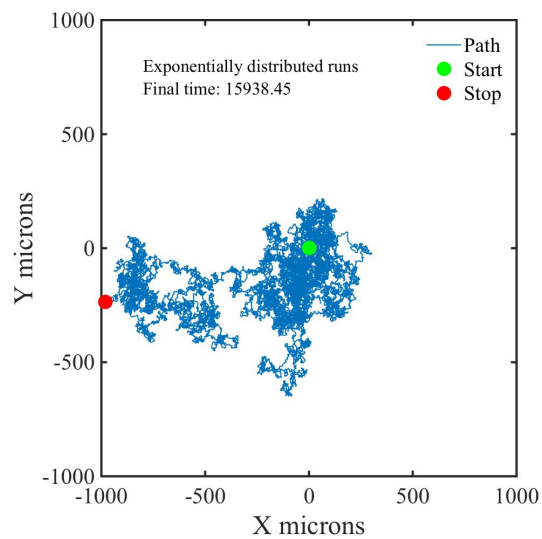
For those more experienced in MATLAB, try and develop a simulation of "diffusion" in which a bacteria swims at a speed of 10 μm/s with each "run" lasting 1 second. In this case the direction of the next run is random. How long will it take, on average, for the bacteria to travel 1mm away from its source? Recall that in class we said that the average time to travel a distance x should be:

$$T = \frac{x^2}{D} \tag{5}$$

What do we mean by *average* here?

If you develop code, can you visualize the results? If you change the length in equation, how does T change? Can you confirm the scaling and estimate D? What happens if the durations of each run is exponentially distributed, so that runs last on average 1 second, but can vary in duration? Did the answer with respect to travel time to reach 1mm change in a quantitative or qualitative way? We will get much further into these ideas later in the class. Here are examples of two runs, one with constant duration and one with exponentially distributed durations:





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